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Input device

Abstract

A Virtual Reality (VR) image generation system includes a sensor; a detection circuit which, in operation, detects a first indicated position indicated by an electronic pen in a first three-dimensional space in accordance with transfer of a signal between the electronic pen and the sensor; one or more processors; and one or more memory devices storing instructions that, when executed by the one or more processors, cause the one or more processors to: detect a first posture of the electronic pen in a second three-dimensional space, the first posture including at least one of a first inclination angle, a first inclination direction and a first rotation angle of the electronic pen, and generate, when the electronic pen is in the first three-dimensional space, a first VR image of drawing by the electronic pen, based on the first indicated position and the first posture.

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Background/Summary

BACKGROUND

Technical Field

(1) The disclosure relates to an input device suitable for use in the case where an electronic pen is used in a space.

Background Art

(2) A drawing system that allows an animation image or the like to be created by performing a drawing by continuously indicating positions via an electronic pen for a coordinate input device called a digitizer is known.

(3) Various methods have been proposed as methods for detecting the inclination and rotation of an electronic pen. Patent Document 1 (Japanese Patent Laid-Open No. 2016-126503) provides an electronic pen capable of detecting the inclination angle and rotation angle thereof with respect to an input surface of a sensor of a position detection device. In the case of this Patent Document 1, signals are exchanged between the electronic pen and the position detection device provided with the sensor through electromagnetic inductive coupling, capacitive coupling, or the like to enable the position detection device to detect the position indicated by the electronic pen, and in addition, the position coordinates of the position indicated by the electronic pen, and the inclination angle and rotation angle of the electronic pen.

(4) Meanwhile, Patent Document 2 (U.S. Pat. No. 9,329,703) describes features in which a sensor capable of measuring the motion and direction of an electronic pen is contained in or attached to the electronic pen, and a detection output of the sensor is transmitted to a position detection device to enable the position detection device to detect the motion and state (e.g., inclination, rotation, etc.) of the electronic pen.

PRIOR ART DOCUMENT

Patent Documents

(5) Patent Document 1: Japanese Patent Laid-Open No. 2016-126503 Patent Document 2: U.S. Pat. No. 9,329,703 B2

SUMMARY

Technical Problems

(6) In the case of Patent Document 1 mentioned above, the position coordinates of the position indicated by the electronic pen and the inclination thereof are detected using a plurality of signal

levels obtained from a plurality of loop coils around the position indicated by the electronic pen. However, with respect to a peripheral region of the sensor, the precision of the position coordinates of the position indicated by the electronic pen will be reduced, and a detection of the inclination of the electronic pen will be difficult, which are problems.

(7) The problems that occur in the case of Patent Document 1 do not occur with the method of Patent Document 2. However, the method of Patent Document 2 has a problem in that a reduced response speed or a reduced precision of position coordinates may occur depending on conditions.

(8) An object of the disclosure is to provide an input device that is able to overcome the above problems.

Technical Solution

(9) In order to solve the above problems, an input device is provided which includes a sensor; a detection circuit which, in operation, detects a first indicated position indicated by an electronic pen in a two-dimensional space in accordance with transfer of a signal between the electronic pen and the sensor; one or more processors; and one or more memory devices storing instructions that, when executed by the one or more processors, cause the one or more processors to: detect a second indicated position indicated by the electronic pen in a three-dimensional space, and generate position information of the electronic pen based on the first indicated position and the second indicated position.

(10) In the input device having the above-described structure, the indicated positions include not only position coordinates but also a posture (e.g., inclination, rotation, etc.). In addition, the input device having the above-described structure allows information regarding a posture included in the position indicated by the electronic pen to be outputted together with position coordinates of the first indicated position, for example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram for explaining an example configuration of a spatial position indication system including an input device according to a first embodiment of the disclosure.

(2) FIG. 2 is a block diagram for explaining an example configuration of various circuits of the spatial position indication system in the example of FIG. 1.

(3) FIG. 3 is a diagram for explaining an input device according to an embodiment of the disclosure.

(4) FIG. 4 is a block diagram for explaining an example configuration of various circuits of an input device according to a second embodiment of the disclosure.

(5) FIGS. 5A and 5B depicts diagrams for explaining an operation of detecting an inclination of an electronic pen through transfer of signals between the electronic pen and a digitizer.

(6) FIG. 6 is a diagram used to explain that the precision with which the position indicated by the electronic pen and the inclination of the electronic pen are detected by the digitizer through transfer of signals between the electronic pen and the digitizer is different between a peripheral region of an input surface and another region.

(7) FIG. 7 is a diagram illustrating a part of a flowchart for explaining a primary operation of the input device according to the second embodiment of the disclosure.

(8) FIG. 8 is a diagram illustrating a part of the flowchart for explaining the primary operation of the input device according to the second embodiment of the disclosure.

(9) FIG. 9 is a diagram for explaining an example configuration of an input device according to a third embodiment of the disclosure.

(10) FIG. 10 is a block diagram for explaining an example configuration of various circuits of the input device according to the third embodiment of the disclosure.

DETAILED DESCRIPTION

(11) Hereinafter, input devices according to embodiments of the disclosure will be described with reference to the accompanying drawings.

First Embodiment

(12) An input device according to an embodiment described below is concerned with a spatial position indication system that includes a digitizer (pen tablet) provided with a sensor that performs at least one of transmission and reception of a signal (hereinafter referred to as “transfer of a signal”) in relation to an electronic pen, and a spatial position detector that detects information regarding a position indicated by the electronic pen in a space. Notice that the position includes not only position coordinates but also a posture (e.g., an inclination, a rotation, etc.) and so on, but in the following description, information regarding the position coordinates and information regarding the posture will be described as separate pieces of information to distinguish both.

(13) In the spatial position indication system according to this embodiment, a display circuit is formed by a head-mounted display, a smart glass, or the like, and a 3D (Three-Dimensional) drawing space is used as a space of a virtual reality (which may be VR (Virtual Reality), MR (Mixed Reality), AR (Augmented Reality), or the like, hereinafter referred to simply as VR). FIG. 1 is a diagram illustrating an outline of the overall configuration of a spatial position indication system including an input device according to a first embodiment, in which a space of a VR is a 3D drawing space. FIG. 2 is a block diagram illustrating an example of the detailed configuration of functions of various circuits of the spatial position indication system including the input device according to the first embodiment.

(14) Specifically, as illustrated in FIG. 1, the spatial position indication system according to this example embodiment includes an electronic pen **10**, a digitizer **20**, a spatial position detector **30**, a spatial drawing information generation device **40**, and a head-mounted display (hereinafter referred to as an HMD) **50**. As illustrated in FIG. 2, in this example, the spatial drawing information generation device **40** has functions of an input information processing circuit **41** and a display image generation circuit **42**, and is formed by, for example, a computer.

(15) The input device according to this first embodiment includes a first detection circuit that detects a position indicated by the electronic pen **10** in accordance with transfer of a signal between the electronic pen **10** and the first detection circuit, a second detection circuit that detects a position indicated by the electronic pen **10** in a three-dimensional space in which the electronic pen **10** exists, and a control circuit that generates position information of the electronic pen **10** on the basis of these indicated positions. In this first embodiment, the first detection circuit is formed by the digitizer **20**, the second detection circuit is formed by the spatial position detector **30** and a spatial information processing circuit **410** of the input processing circuit **41**, and the control circuit is formed by the input information processing circuit.

(16) An electronic pen that employs an electromagnetic induction method is used as an example of the electronic pen **10** in this first embodiment, but it is to be understood that the electronic pen **10** may employ a capacitive coupling method or the like instead of the electromagnetic induction method. The digitizer **20** includes a slim housing **21** in the shape of a rectangular parallelepiped, and a front surface thereof is an input surface **21S** for a position indication by the electronic pen **10**. In addition, the digitizer **20** includes a sensor **22** and a position detection circuit **23** (see FIG. 2).

(17) Although not illustrated in the figures, the sensor **22** includes a plurality of loop coils arranged in a horizontal direction (i.e., an X-axis direction) of the housing of the digitizer **20** and in a vertical direction (i.e., a Y-axis direction) of the housing. In this example, the digitizer **20** is configured to employ the electromagnetic induction method in accordance with the electronic pen **10**, but may alternatively employ a method other than the electromagnetic induction method.

(18) Although not illustrated in the figures, the electronic pen **10** includes, at a pen point, a resonant circuit (not illustrated) made up of a coil and a capacitor, and transfer of signals is performed between the electronic pen **10** and the sensor **22** of the digitizer **20** through electromagnetic

inductive coupling between the loop coils of the sensor **22** of the digitizer **20** and the resonant circuit of the electronic pen **10**.

(19) The position detection circuit **23** of the digitizer **20** supplies signals to the electronic pen **10** through the loop coils of the sensor **22**, and receives signals fed back from the electronic pen **10** through the loop coils, and, on the basis of the signals received, detects a position indicated by the electronic pen **10** in a detection region of the sensor **22**. Notice that, in this embodiment, the digitizer **20** is configured to detect a position indicated by the pen point of the electronic pen **10** as the position indicated by the electronic pen **10**.

(20) In the digitizer **20** in this example, the plurality of loop coils of the sensor **22** are arranged to cover substantially the entire area of the input surface **21S**.

(21) In addition, in this embodiment, a position detection region, in which the position indicated by the electronic pen **10** can be detected by the digitizer **20**, includes not only a planer region when the pen point of the electronic pen **10** is in contact with the input surface **21S** of the digitizer **20**, but also a spatial region (i.e., a hover region for the electronic pen **10** in a hovering state) in which the position indicated by the electronic pen **10** can be detected through transfer of signals through electromagnetic coupling when the pen point of the electronic pen **10** is not in contact with the input surface **21S** of the digitizer **20** and is spaced apart from the input surface **21S** in a direction (i.e., a Z-axis direction perpendicular to both the X-axis direction and the Y-axis direction) perpendicular to the input surface **21S**.

(22) In this example, the spatial position detector **30** is configured to set a three-dimensional spatial region in which the digitizer **20** exists, and to be capable of, in the three-dimensional spatial region, detecting the position indicated by the pen point of the electronic pen **10** and the posture (referred to as a pen posture) of the electronic pen **10**, such as the inclination angle and the rotation angle of the electronic pen **10**, and also detecting the position of the digitizer **20** and the inclination angle and inclination direction thereof with respect to a horizontal plane.

(23) The spatial position detector **30** is configured to include two light emitting/tracking devices **31A** and **31B**, and a plurality of optical position notification circuits (hereinafter referred to as trackers) **32A**, **32B**, **32C**, and **32D**. In this embodiment, as described below, on the HMD **50**, a 3D drawing image drawn in a spatial motion detection region MD including a position detection region DT of the digitizer **20** is displayed as a virtual display image, and a virtual display image of the electronic pen **10** is displayed.

(24) The two light emitting/tracking devices **31A** and **31B** have the same configuration, and each include a laser beam emitting circuit that emits infrared laser beams, searching means that searches the spatial motion detection region MD with the emitted infrared laser beams, and optical position detection means that detects a light emission of a light emitting circuit of each of the trackers **32A**, **32B**, **32C**, and **32D** in response to receiving the infrared laser beam.

(25) In this embodiment, objects (i.e., objects to be searched for in a spatial search region of the spatial position detector **30**) onto which the trackers **32A**, **32B**, **32C**, and **32D** are attached are the electronic pen **10** and the digitizer **20** as mentioned above. Specifically, in this example, the tracker **32A** and the tracker **32B** are attached onto a top left corner and a bottom right corner, respectively, of the slim housing of the digitizer **20** in the shape of a rectangular parallelepiped, to enable notification of the position of the digitizer **20** and the inclination angle and inclination direction thereof with respect to the horizontal plane. In addition, the tracker **32C** is attached onto the electronic pen **10** on a pen-point side thereof, and the tracker **32D** is attached onto the electronic pen **10** on a rear-end side thereof opposite to the pen-point side in a direction of an axis-center of a housing of the electronic pen **10**, to enable notification of the position of the electronic pen **10** and the pen posture (e.g., the inclination angle and the rotation angle).

(26) Each of the light emitting/tracking devices **31A** and **31B** controls the laser beam emitting circuit via the searching means, and performs a search by emitting the infrared laser beams so as to search and scan the spatial motion detection region MD to detect tracker positions. Each of the

trackers **32A**, **32B**, **32C**, and **32D** performs monitoring for a reception of the infrared laser beam with a sensor, and, when a reception of the infrared laser beam is detected by the sensor, lights up the light emitting circuit, formed by an LED (Light Emitting Diode).

(27) The light emitting/tracking devices **31A** and **31B** detect the positions of the objects onto which the trackers **32A**, **32B**, **32C**, and **32D** are attached in the spatial motion detection region MD, by detecting the light emission of the light emitting circuits of the trackers **32A**, **32B**, **32C**, and **32D**. Each of the light emitting/tracking devices **31A** and **31B** is configured to be capable of, when a light emission of the light emitting circuit of each of the trackers **32A**, **32B**, **32C**, and **32D** has been detected, detecting a time that has elapsed from an emission time of the emitted infrared laser beam at the time of the detection as well. In this case, each of the trackers **32A**, **32B**, **32C**, and **32D** performs a different light emission in accordance with its own identification information.

(28) Each of the two light emitting/tracking devices **31A** and **31B** is connected to the spatial drawing information generation device **40** in a wired or wireless manner, and provides, to the spatial drawing information generation device **40**, information regarding the spatial position of each of the trackers **32A**, **32B**, **32C**, and **32D** detected in the spatial motion detection region MD.

(29) The information regarding the spatial position of each of the trackers **32A**, **32B**, **32C**, and **32D** in the spatial motion detection region MD detected by the two light emitting/tracking devices **31A** and **31B** is, in this example, supplied to the spatial information processing circuit **410** of the input information processing circuit **41** of the spatial drawing information generation device **40** as illustrated in FIG. 2.

(30) In this example, the spatial information processing circuit **410** includes a spatial position detection circuit **4101**, a pen posture detection circuit **4102**, and a digitizer posture detection circuit **4103**. The spatial position detection circuit **4101** detects the position of the digitizer **20** in the spatial motion detection region MD from the trackers **32A** and **32B**, and also detects the position indicated by the electronic pen **10** as the position of the pen point of the electronic pen **10**, and the position of a rear end of the electronic pen **10**, from the trackers **32C** and **32D**. In this example, the pen posture detection circuit **4102** detects the pen posture, including the inclination angle and the rotation angle of the electronic pen **10**, from the trackers **32C** and **32D**. In addition, the digitizer posture detection circuit **4103** detects the position indicated by the electronic pen **10** as the position of the point.

(31) In addition, in this embodiment, the input information processing circuit **41** generates information to be supplied to the display image generation circuit **42** from the information regarding the position indicated by the electronic pen **10** in the position detection region DT, which has been detected by the digitizer **20**, and the information regarding the position (i.e., the indicated position and the position of the rear end) of the electronic pen **10** in the spatial motion detection region MD, pen posture information, and digitizer posture information, which have been detected by the spatial information processing circuit **410** of the input information processing circuit **41**. Then, the input information processing circuit **41** supplies the generated information to the display image generation circuit **42**.

(32) In addition, as illustrated in FIG. 2, in this embodiment, the display image generation circuit **42** of the spatial drawing information generation device **40** includes a drawing image generation circuit **421** for generating a 3D drawing image, and a VR image generation circuit **422** for generating VR images to be displayed on the HMD **50**.

(33) The drawing image generation circuit **421** generates a 3D drawing image on the basis of a position indication by the electronic pen **10**, and also performs a process of, for example, transforming, rotating, or moving a 3D drawing image on the basis of a gesture made by an operator of the electronic pen **10**. In addition, in this embodiment, the drawing image generation circuit **421** causes the pen posture, including the inclination angle, inclination direction, and rotation angle of the electronic pen **10** with respect to the input surface **21S** of the digitizer **20**, as well to be reflected in the generation of the 3D drawing image. For example, in the case where

transfer of signals between the electronic pen **10** and the digitizer **20** is performed, the inclination angle and inclination direction of the electronic pen **10** with respect to the input surface **21S** of the digitizer **20** are used. Hereinafter, a process related to the 3D drawing image will be referred to as a 3D drawing-related process.

(34) In this embodiment, the pen posture, including the inclination angle, inclination direction, and rotation angle of the electronic pen **10**, can be detected through the spatial position detector **30** using the trackers **32C** and **32D** attached onto the electronic pen **10**. Thus, in this embodiment, the input information processing circuit **41** is configured to supply, to the display image generation circuit **42**, information regarding the pen posture of the electronic pen **10** detected by the spatial position detector **30** as information to be used in the 3D drawing-related process.

(35) Specifically, as described below, the input information processing circuit **41** is configured to supply, to the display image generation circuit **42**, the information (i.e., information regarding the position of the pen point) about the position indication by the electronic pen **10**, and the information regarding the pen posture of the electronic pen **10**, to cause information regarding the pen posture, such as the inclination, rotation of the electronic pen **10** at the time of a drawing input based on the position indication by the electronic pen **10** to be reflected in the 3D drawing image.

(36) In addition, as illustrated in FIG. 2, the display image generation circuit **42** includes a gesture detection processing circuit **423** for detecting a gesture performed by the operator of the electronic pen **10**. In addition, in this embodiment, the input information processing circuit **41** is configured to supply, to the gesture detection processing circuit **423**, the information regarding the pen posture, and position information (corresponding to the information regarding the indicated position) of the pen point of the electronic pen **10** detected in the spatial position detector **30**.

(37) As described above, in this embodiment, not only the information regarding the position indicated by the electronic pen **10** detected in the digitizer **20**, but also the information regarding the position (i.e., the position of the pen point) indicated by the electronic pen **10** and the information regarding the pen posture detected in the spatial position detector **30**, are used as the information to be used in the 3D drawing-related process.

(38) The VR image generation circuit **422** of the display image generation circuit **42** generates the VR images to be displayed on the HMD **50**. In this embodiment, examples of the VR images include a VR image of the electronic pen **10** and a VR image of the 3D drawing image generated by the drawing image generation circuit **421**, and the VR image of the 3D drawing image generated by the drawing image generation circuit **421** and the VR image of the electronic pen **10** are displayed in a 3D fashion on the HMD **50**. Note that the VR image generation circuit **422** may be configured to generate a VR image of the digitizer **20** as well. Hereinafter, a process related to the generation of a VR image in the VR image generation circuit **422** will be referred to as a VR image-related process.

(39) In this embodiment, information for generating the VR image of the electronic pen **10** is supplied from the input information processing circuit **41** to the VR image generation circuit **422**, and information of the generated 3D drawing image is supplied from the drawing image generation circuit **421** to the VR image generation circuit **422**. Then, VR image information generated in the VR image generation circuit **422** is supplied to the HMD **50** through a display drive circuit **424**, and is displayed on a display screen, which is, for example, formed by an LCD (Liquid Crystal Display), of the HMD **50**.

(40) In this case, when the electronic pen **10** exists in the position detection region DT, the information regarding the position indicated by the electronic pen from the digitizer **20**, which is capable of detecting the position (i.e., the position of the pen point) of the electronic pen **10** with higher precision than the spatial position detector **30**, and the information regarding the pen posture, such as the inclination, rotation detected using the spatial position detector **30** are supplied from the input information processing circuit **41** to the VR image generation circuit **422** of the display image generation circuit **42**.

(41) Meanwhile, when the electronic pen **10** exists in the spatial motion detection region MD, the digitizer **20** is not able to detect the position indicated by the electronic pen **10**, and accordingly, the information regarding the pen posture and the information regarding the position indicated by the electronic pen detected using the spatial position detector **30** are supplied from the input information processing circuit **41** to the VR image generation circuit **422** of the display image generation circuit **42**.

(42) As described above, in this embodiment, not only the information regarding the pen posture and the information regarding the position (i.e., the position of the pen point) indicated by the electronic pen **10** detected by the spatial position detector **30**, but also the information regarding the position indicated by the electronic pen **10** detected by the digitizer **20**, are used as information to be used in the VR image-related process.

(43) The input information processing circuit **41** generates the information to be used in the drawing-related process as described above and the information to be used in the VR image-related process from the information from the digitizer **20** and the information from the spatial position detector **30**, and supplies the generated information to the display image generation circuit **42**, and in addition, generates a switching control signal SE for selection control depending on whether the electronic pen **10** exists in the position detection region DT or in the spatial motion detection region MD.

(44) As described above, in this embodiment, a complementary use of information regarding a spatial coordinate system of the position detection region DT of the digitizer **20** and information regarding a spatial coordinate system of the spatial motion detection region MD of the spatial position detector **30** is possible in each of the 3D drawing-related process and the VR image-related process. In this case, each of the two spatial coordinate systems may be independently set, but in this embodiment, the information regarding one of the two spatial coordinate systems is converted to information regarding the other spatial coordinate system to enable the information to be dealt with as information in a common coordinate space. In this embodiment, the information regarding the spatial coordinate system of the spatial motion detection region MD of the spatial position detector **30** is converted to information regarding the spatial coordinate system of the position detection region DT of the digitizer **20**.

(45) In the spatial position indication system according to this embodiment, the information regarding the pen posture of the electronic pen **10** detected in the spatial motion detection region MD of the spatial position detector **30** can be used in both the 3D drawing-related process and the VR image-related process.

(46) In this case, in the VR image-related process, the information regarding the pen posture of the electronic pen **10** is reflected in the posture of the VR image of the electronic pen **10**. This information regarding the pen posture of the electronic pen **10** includes information regarding the inclination angle of the electronic pen **10**, and in the 3D drawing-related process, this inclination angle of the electronic pen **10** is an inclination angle relative to the input surface of the sensor **22** of the digitizer **20**. Meanwhile, the information regarding the pen posture of the electronic pen **10** detected in the spatial motion detection region MD of the spatial position detector **30** represents an inclination angle in the spatial motion detection region MD of the spatial position detector **30** with respect to the direction of gravity of the earth or a horizontal plane, for example.

(47) If the digitizer **20** can be set such that a direction perpendicular to the input surface **21S** of the sensor **22** thereof precisely coincides with the Z-axis direction of the spatial motion detection region MD of the spatial position detector **30**, the inclination angle of the electronic pen **10** detected in the space of the spatial motion detection region MD of the spatial position detector **30** will coincide with the inclination angle relative to the input surface of the sensor **22** of the digitizer **20**. In actuality, however, the digitizer **20** is sometimes set at an angle with respect to the spatial motion detection region MD of the spatial position detector **30**.

(48) Accordingly, in this embodiment, the absolute inclination angle of the electronic pen **10**

detected in the space of the spatial motion detection region MD of the spatial position detector **30** is converted to the inclination angle relative to the input surface of the sensor **22** of the digitizer **20** as described below.

(49) FIG. **3** is a diagram used for explaining a process of converting the inclination angle. FIG. **3** illustrates a spherical coordinate system with the pen point of the electronic pen **10** at a point **O** in the center of a sphere. An X-Y plane (i.e., a plane including the X-axis direction and the Y-axis direction perpendicular to each other) in FIG. **3** corresponds to a horizontal plane in the spatial region of the spatial motion detection region MD of the spatial position detector **30**.

(50) In FIG. **3**, a point representing the position of an edge of a rear end portion of the housing of the electronic pen **10**, which is on the opposite side with respect to the pen point in the direction of the axis-center, is expressed by spherical coordinates. In this case, the radius r of the sphere corresponds to the length of the housing of the electronic pen **10** in the direction of the axis-center thereof. The inclination angle and the inclination direction of the electronic pen **10** in the space of the spatial motion detection region MD when the edge of the rear end portion in the direction of the axis-center of the housing of the electronic pen **10** is at a point P in the spherical coordinate system of FIG. **3** can be detected as θ and α , respectively, by using the spatial position detector **30**. In addition, in the example of FIG. **3**, the inclination angle and the inclination direction of the housing of the digitizer **20** in the space of the spatial motion detection region MD can be detected as δ' and α' , respectively, by using the spatial position detector **30**.

(51) Then, as illustrated in FIG. **3**, assuming that the position of the rear-end edge of the electronic pen **10** in a state in which the direction of the axis-center of the housing of the electronic pen **10** coincides with a vertical direction in the space of the spatial motion detection region MD is at a point Q, the position of the rear-end edge of the electronic pen **10** when the direction of the axis-center of the housing of the electronic pen **10** coincides with the direction perpendicular to the input surface **21S** of the sensor **22** of the digitizer **20** is at a point R.

(52) Accordingly, if an angle formed between a spatial vector OR from the point O to the point R and a spatial vector OP from the point O to the point P is obtained, use of this angle makes it possible to convert the inclination angle and the inclination direction of the electronic pen **10** detected using the spatial position detector **30** to the inclination angle and the inclination direction of the electronic pen **10** relative to the input surface **21S** of the sensor **22** of the digitizer **20**.

(53) Note that, with the reverse of the above-described conversion, the inclination angle and the inclination direction of the electronic pen **10** relative to the input surface **21S** of the sensor **22** of the digitizer **20** can be converted to the inclination angle and the inclination direction of the electronic pen **10** detected using the spatial position detector **30**.

(54) Next, an example configuration of the input information processing circuit **41** of FIG. **2**, which is configured to implement the above-described features, will be described below. That is, the position detection circuit **23** of the digitizer **20** supplies a detection output of the position indicated by the electronic pen **10** to a selection circuit **411** as one input signal therefor, and supplies the detection output to a selection circuit **412** as one input signal therefor. Notice that this information supplied from the position detection circuit **23** to each of the selection circuits **411** and **412** includes, in addition to the detection output of the position indicated by the electronic pen **10**, information regarding a pen pressure applied to the electronic pen **10**.

(55) Meanwhile, the spatial position detection circuit **4101** of the spatial information processing circuit **410** supplies a detection output of the spatial position (i.e., the position indicated by the electronic pen) of the pen point of the electronic pen **10** to a coordinate conversion circuit **413**. This coordinate conversion circuit **413** converts the information regarding the spatial coordinate system of the spatial motion detection region MD of the spatial position detector **30** to information regarding the spatial coordinate system of the position detection region DT of the digitizer **20**. Then, the coordinate conversion circuit **413** supplies the converted coordinate output to each of the selection circuits **411** and **412** as another input signal therefor.

(56) In addition, in this embodiment, the position detection circuit **23** of the digitizer **20** supplies, to a selection control signal generation circuit **414**, information regarding the signal level of a reception signal from the electronic pen **10**. The selection control signal generation circuit **414** detects a distance by which the pen point of the electronic pen **10** is spaced from the input surface **21S** of the digitizer **20** from the signal level of the reception signal from the electronic pen **10**, and generates a selection control signal SE on the basis of the detected distance from the input surface **21S**.

(57) In this case, the selection control signal SE is a signal for controlling the detection output of the position from the position detection circuit **23** of the digitizer **20** to be selected when the distance by which the pen point of the electronic pen **10** is spaced from the input surface **21S** of the digitizer **20** is equal to or smaller than a maximum height Lz in the Z-axis direction that allows the digitizer **20** to detect the hovering state of the electronic pen, and controlling the detection output from the spatial position detection circuit **4101** to be selected when the aforementioned distance is greater than the aforementioned maximum height Lz.

(58) The selection circuit **411** selectively supplies one of the inputs to an associating circuit **415** for the 3D drawing-related process in accordance with the selection control signal SE. Meanwhile, the selection circuit **412** selectively supplies one of the inputs to an associating circuit **416** for the VR image-related process in accordance with the selection control signal SE.

(59) Although, in the example of FIG. **2**, the two selection circuits **411** and **412** are provided for the 3D drawing-related process and the VR image-related process, respectively, to clearly distinguish between the two processes, only one common selection circuit would suffice in reality, and an output from this common selection circuit may be supplied to each of the associating circuits **415** and **416**.

(60) The information regarding the pen posture detected by the pen posture detection circuit **4102** of the spatial information processing circuit **410** with use of the spatial position detector **30** is, as it is, supplied to the associating circuit **416** for the VR image-related process.

(61) The associating circuit **416** for the VR image-related process associates the information regarding the position indicated by the electronic pen **10** from the selection circuit **412** with the information regarding the pen posture from the pen posture detection circuit **4102** of the spatial information processing circuit **410** to form a pair, and supplies the pair to the VR image generation circuit **422** of the display image generation circuit **42**. In this case, the associating in the associating circuit **416** means associating, with the position indicated by the electronic pen **10** outputted from the selection circuit **412**, the information regarding the pen posture detected at the position indicated by the electronic pen **10** to form a pair.

(62) In the case where the information regarding the position indicated by the electronic pen **10** and the information regarding the pen posture associated with each other in the associating circuit **416** are both information from the spatial information processing circuit **410**, the associating can be accomplished in the associating circuit **416** by simply pairing pieces of information of the same timing with output timing of these pieces of information being adjusted in the spatial information processing circuit **410**.

(63) In contrast, in the case where the information regarding the position indicated by the electronic pen **10** subjected to the associating in the associating circuit **416** is information from the digitizer **20**, the information regarding the position indicated by the electronic pen **10** and the information regarding the pen posture are associated with each other in the associating circuit **416** with a lag (i.e., a processing delay) in output timing between the detection results of the digitizer **20** and the spatial position detector **30** taken into account.

(64) The VR image generation circuit **422** of the display image generation circuit **42** generates the VR image of the electronic pen **10** using the information regarding the position indicated by the electronic pen **10** and the information regarding the pen posture from the associating circuit **416**. In this case, the VR image of the electronic pen **10** is generated to be displayed at a position based on

the information regarding the position indicated by the electronic pen **10**.

(65) In this case, while the information regarding the position indicated by the electronic pen **10** from the selection circuit **412** is selectively chosen in accordance with the selection control signal SE, the information regarding the indicated position from the spatial position detection circuit **4101** is subjected to coordinate conversion in the coordinate conversion circuit **413** so that the digitizer **20** and the spatial position detector **30** share a common coordinate space, and therefore, a switching at the selection circuit **412** would not cause a faulty display, such as, for example, a jumping movement of the position at which the VR image of the electronic pen **10** is displayed.

(66) The information regarding the pen posture from the pen posture detection circuit **4102** of the spatial information processing circuit **410** is also supplied to the associating circuit **415** for the 3D drawing-related process through a posture conversion circuit **417**. The information regarding the pen posture from the pen posture detection circuit **4102** and the information regarding the posture of the digitizer **20** detected in the digitizer posture detection circuit **4103** are supplied to the posture conversion circuit **417**, and in the posture conversion circuit **417**, a process of converting the pen posture of the electronic pen **10** in the spatial coordinate system detected in the spatial position detector **30** to a pen posture relative to the input surface **21S** of the sensor **22** of the digitizer **20** as described above with reference to FIG. **3** is performed.

(67) The associating circuit **415** for the 3D drawing-related process associates the information regarding the position indicated by the electronic pen **10**, which has been supplied from the selection circuit **411**, and the information regarding the pen posture detected in the pen posture detection circuit **4102** converted to information regarding the pen posture relative to the input surface **21S** of the digitizer **20**, which has been supplied from the posture conversion circuit **417**, with each other to form a pair, and supplies the pair to each of the drawing image generation circuit **421** and the gesture detection processing circuit **423** of the display image generation circuit **42**. In this case, as is similarly the case with the associating circuit **416**, the associating in the associating circuit **415** means associating, with the position indicated by the electronic pen **10** outputted from the selection circuit **412**, the information regarding the pen posture detected at the position indicated by the electronic pen **10** to form a pair.

(68) The drawing image generation circuit **421** has a pen drawing function of drawing a precise line or the like on the basis of the detection output of the position indicated by the electronic pen **10** from the digitizer **20** and the information regarding the pen posture detected in the spatial position detector **30** in this embodiment, and a gesture processing function of performing a drawing process based on a motion (gesture) detected in the gesture detection processing circuit **423** on the basis of the pen posture and the spatial position of the electronic pen **10** detected by the spatial position detector **30**.

(69) In addition, the selection control signal SE is supplied from the selection control signal generation circuit **414** to each of the drawing image generation circuit **421** and the gesture detection processing circuit **423**, and the gesture detection processing circuit **423** is controlled to operate when the position indicated by the electronic pen **10** lies outside of the spatial region of the position detection region DT of the digitizer **20**, while the drawing image generation circuit **421** is selectively controlled to perform the pen drawing function when the position indicated by the electronic pen **10** lies within the spatial region of the position detection region DT of the digitizer **20**, and perform a gesture process when the position indicated by the electronic pen **10** lies outside of the spatial region of the position detection region DT of the digitizer **20**.

(70) In this case, in this embodiment, the coordinate conversion is performed using the coordinate conversion circuit **413** so that the digitizer **20** and the spatial position detector **30** share a common coordinate space, and therefore, a switching in the position indicated by the electronic pen **10** at the selection circuit **411** would not cause a jumping displacement of the position indicated by the electronic pen **10**. Moreover, even the pen posture detected by the spatial position detector **30** is appropriately reflected in the 3D drawing image because the pen posture is converted to the pen

posture relative to the input surface 21S of the digitizer 20.

(71) As described above, 3D drawing image information generated in the drawing image generation circuit 421 is supplied to the VR image generation circuit 422 to form a VR image, and the VR image is supplied to the HMD 50 through the display drive circuit 424 to be displayed. In this case, in this embodiment, the spatial coordinate system of the 3D drawing image and the spatial coordinate system of the VR image, such as of the electronic pen 10, are identical because the positions indicated by the electronic pen 10 are expressed by spatial coordinates common to the digitizer 20 and the spatial position detector 30 through use of the coordinate conversion circuit 413. Accordingly, the positions indicated by the electronic pen 10 for the 3D drawing image are not different between the 3D drawing-related process and the VR image-related process, eliminating the need for the VR image generation circuit 422 to correct spatial coordinates for the two.

(72) As will be understood from the foregoing description, this first embodiment produces a beneficial effect of being able to use the pen posture of the electronic pen 10 detected in the spatial position detector 30 in the generation of the 3D drawing image in the 3D drawing-related process, without the need for the electronic pen 10 and the digitizer 20 to have a function of detecting the pen posture, such as the inclination angle, rotation angle of the electronic pen 10. In addition, this first embodiment also produces a beneficial effect of being able to use the information regarding the pen posture of the electronic pen 10 detected by the spatial position detector 30 in the spatial region outside of the position detection region DT of the digitizer 20 for a gesture detection to generate the 3D drawing image. Further, this first embodiment is characterized in that a switching between the output from the digitizer 20 and the output from the spatial position detector 30 in the 3D drawing-related process would not cause a coordinate displacement, since spatial coordinates in the detection region of the digitizer 20 and spatial coordinates in the detection region of the spatial position detector 30 can be handled as common coordinates.

(73) Accordingly, the operator is able to seamlessly shift from detailed drawing to an operation by a gesture by simply moving the electronic pen 10 in a space over the digitizer 20 without being conscious of a switching between the digitizer 20 and the spatial position detector 30.

(74) In addition, in the above-described first embodiment, as the position information (i.e., the position information of the pen point) of the electronic pen 10 used to generate the VR image of the electronic pen 10 in the VR image-related process, the position detection output of the digitizer 20, which provides a higher precision and a higher response speed than the detection output of the spatial position detector 30, can be used in the spatial region of the position detection region DT of the digitizer 20, and accordingly, a VR image that is properly responsive to an actual operation of the electronic pen by the user can be obtained.

(75) It is needless to say that, since the above-described spatial drawing information generation device 40 is formed by a computer, the various circuits of the input information processing circuit 41 and the display image generation circuit 42 can be formed as functional software circuits implemented by a software program. For example, the various circuits described herein are implemented by one or more processors and one or more memory devices storing instructions that, when executed by the one or more processors, cause the circuits to perform the acts described herein.

Second Embodiment

(76) In the above-described first embodiment, each of the electronic pen 10 and the digitizer 20 does not have the function of detecting the pen posture, such as the inclination angle, rotation angle of the electronic pen, but each of the electronic pen 10 and the digitizer 20 may alternatively be configured to have the function of detecting the pen posture, such as the inclination angle, rotation angle of the electronic pen.

(77) An input device according to a second embodiment, which will be described below, is used in a system similar to the spatial position indication system described above with respect to the first embodiment, and includes an electronic pen and a digitizer having the function of detecting the pen

posture, such as the inclination angle, rotation angle of the electronic pen.

(78) FIG. 4 illustrates an example configuration of important circuits of a spatial position indication system according to this second embodiment. The spatial position indication system according to this second embodiment has a configuration similar to that of the spatial position indication system according to the first embodiment, but is provided with an electronic pen **10A** (not illustrated in FIG. 4) and a digitizer **20A** having a configuration for detecting the pen posture in place of the electronic pen **10** and the digitizer **20**. In addition, in place of the input information processing circuit **41** of the spatial drawing information generation device **40** in the first embodiment, an input information processing circuit **41A** having a configuration illustrated in FIG. 4 is provided.

(79) The spatial position indication system according to this second embodiment is also provided with the spatial position detector **30**, the display image generation circuit **42**, and the HMD **50**, which have been described in connection with the first embodiment, and are not illustrated in FIG. 4 as no significant changes therein are made from the first embodiment. In addition, circuits of the digitizer **20A** and the input information processing circuit **41A** illustrated in FIG. 4 which have their equivalents in the digitizer **20** and the input information processing circuit **41** illustrated in FIG. 2 are designated by the same reference numerals as those of their equivalents, and detailed descriptions thereof will be omitted.

(80) As illustrated in FIG. 5A, when the electronic pen **10A** is perpendicular to the input surface of the sensor, the signal levels of a specific number (which is greater than one) of, e.g., seven, loop coil positions X_{i-3} , X_{i-2} , X_{i-1} , X_i , X_{i+1} , X_{i+2} , and X_{i+3} assume a symmetric shape with the indicated position in the center.

(81) In contrast, as illustrated in FIG. 5B, when the electronic pen **10A** is inclined, the signal levels of these seven loop coil positions X_{i-3} , X_{i-2} , X_{i-1} , X_i , X_{i+1} , X_{i+2} , and X_{i+3} vary in accordance with the direction in which the electronic pen **10A** is inclined and the angle at which the electronic pen **10A** is inclined. The position coordinates of the position indicated by the electronic pen **10A** and the inclination angle and inclination direction of the electronic pen **10A** can be detected on the basis of the specific number of signal levels.

(82) However, with respect to an input surface (i.e., a detection region) INS of the sensor illustrated in FIG. 6, only a smaller number (which is greater than one) of signal levels than the aforementioned specific number can be obtained in, for example, a peripheral region PE indicated by oblique lines outside of the position of a dotted line. Accordingly, the following point is taken into account, in this second embodiment, the precision with which the position (i.e., the position of the pen point) indicated by the electronic pen **10A** and the pen posture of the electronic pen **10A** can be detected in the peripheral region PE of the digitizer **20A** is reduced, and in the case of this second embodiment, the detection precision of the position (i.e., the position of the pen point) indicated by the electronic pen **10A** and the pen posture of the electronic pen **10A** detected in the spatial position detector **30** is higher.

(83) That is, in this second embodiment, even when the position of the pen point of the electronic pen **10A** is in the spatial region of the position detection region DT, in which the position indicated by the electronic pen **10A** and the pen posture can be detected in the digitizer **20A**, information regarding the position indicated by the electronic pen **10A** and the pen posture detected in the spatial position detector **30** is used, instead of information regarding the position indicated by the electronic pen **10A** and the pen posture detected in the digitizer **20A**, if the position of the pen point of the electronic pen **10A** is in the peripheral region PE or in a spatial region over the peripheral region PE.

(84) As illustrated in FIG. 4, the digitizer **20A** according to this second embodiment includes a posture detection circuit **24**. This posture detection circuit **24** detects the pen posture, such as the inclination angle, inclination direction, rotation angle of the electronic pen **10A** on the basis of transfer of signals between the electronic pen **10A** (not illustrated) and a sensor **22** of the digitizer **20A**. Information about the pen posture of the electronic pen **10A** detected in this posture detection

circuit **24** is supplied to one input terminal of a selection circuit **418** of the input information processing circuit **41A**. Information about the pen posture of the electronic pen **10A** detected using the spatial position detector **30** is supplied from a pen posture detection circuit **4102** of the spatial information processing circuit **410** to another input terminal of the selection circuit **418**.

(85) The information regarding the pen posture of the electronic pen **10A** is supplied from the selection circuit **418** to an associating circuit **415**, and is subjected to an associating process similar to the above-described associating process to be associated with the information regarding the position indicated by the electronic pen **10A** from a selection circuit **411**, and the resultant is supplied to each of the drawing image generation circuit **421** and the gesture detection processing circuit **423** of the display image generation circuit **42**.

(86) Meanwhile, in this second embodiment, the information regarding the pen posture of the electronic pen **10A** detected in the pen posture detection circuit **4102** of the spatial information processing circuit **410** is supplied to one input terminal of a selection circuit **419**. In addition, the information regarding the pen posture of the electronic pen **10A** is supplied from the posture detection circuit **24** of the digitizer **20A** to a posture conversion circuit **417R**, while information regarding a digitizer posture including the inclination angle and inclination direction of the digitizer **20A** detected in a digitizer posture detection circuit **4103** of the spatial information processing circuit **410** is supplied to the posture conversion circuit **417R**.

(87) In the posture conversion circuit **417R**, the information regarding the pen posture of the electronic pen **10A** from the posture detection circuit **24** of the digitizer **20A**, and the information regarding the posture of the digitizer **20A** from the digitizer posture detection circuit **4103** of the spatial information processing circuit **410**, are subjected to the reverse of the conversion performed in a posture conversion circuit **417**. That is, the pen posture of the electronic pen **10A** relative to the input surface **21S** detected in the digitizer **20A** is converted to an absolute posture of the digitizer **20A** in the spatial coordinate system of the spatial motion detection region MD to be detected in the spatial position detector **30**.

(88) Then, information regarding the pen posture of the electronic pen **10A** resulting from the posture conversion in the posture conversion circuit **417R** is supplied to another input terminal of the selection circuit **419**. Information about the pen posture of the electronic pen **10A** to be selected in and outputted from the selection circuit **419** as described below is supplied to an associating circuit **416**, and is subjected to an associating process similar to the above-described associating process to be associated with the information regarding the position indicated by the electronic pen **10A** from a selection circuit **412**, and the resultant is supplied to the VR image generation circuit **422** of the display image generation circuit **42**.

(89) A selection control signal generation circuit **414A** according to this second embodiment generates selection control signals SEA for the selection circuit **411**, the selection circuit **412**, the selection circuit **418**, and the selection circuit **419**. In addition, the selection control signal generation circuit **414A** generates a selection control signal SE similar to the selection control signal SE generated by the selection control signal generation circuit **414** according to the above-described first embodiment, and thus selectively controls processes performed by the drawing image generation circuit **421** and the gesture detection processing circuit **423** of the display image generation circuit **42** in a manner similar to that of the above-described first embodiment.

(90) The selection control signals SEA from the selection control signal generation circuit **414A** according to this second embodiment are configured to not only selectively control the selection circuits **411**, **412**, **418**, and **419** depending on whether the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies within or outside of the spatial region of the position detection region DT of the digitizer **20A**, but also selectively control the selection circuits **411**, **412**, **418**, and **419** depending on whether the position lies in the peripheral region PE (see FIG. 6) of the input surface **21S** of the digitizer **20A** or in a central region inside of the peripheral region PE of the input surface **21S**.

(91) Accordingly, in addition to information regarding the signal level of the reception signal from the electronic pen **10A**, the information regarding the position indicated by the electronic pen **10A** is supplied from a position detection circuit **23** of the digitizer **20A** to the selection control signal generation circuit **414A** according to this second embodiment. Then, the selection control signal generation circuit **414A** generates the selection control signals SEA for selectively controlling the selection circuits **411**, **412**, **418**, and **419** depending on whether the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies within the spatial region of the position detection region DT and in the spatial region of the central region inside of the peripheral region PE or in a spatial region outside thereof.

(92) Through the selection control signal SEA, the selection circuit **411** selects the information regarding the position indicated by the electronic pen **10A** from the position detection circuit **23** of the digitizer **20A** and supplies the selected information to the associating circuit **415** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies within the spatial region of the position detection region DT and in the spatial region of the central region inside of the peripheral region PE, and selects the information regarding the position indicated by the electronic pen **10A** from a spatial position detection circuit **4101** and a coordinate conversion circuit **413** and supplies the selected information to the associating circuit **415** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies in a spatial region outside thereof.

(93) Meanwhile, through the selection control signal SEA, the selection circuit **412** selects the information regarding the position indicated by the electronic pen **10A** from the position detection circuit **23** of the digitizer **20A** and supplies the selected information to the associating circuit **416** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies within the spatial region of the position detection region DT and in the spatial region of the central region inside of the peripheral region PE, and selects the information regarding the position indicated by the electronic pen **10A** from the spatial position detection circuit **4101** subjected to the coordinate conversion process in the coordinate conversion circuit **413** and supplies the selected information to the associating circuit **416** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies in a spatial region outside thereof.

(94) Meanwhile, through the selection control signal SEA, the selection circuit **418** selects the information regarding the pen posture of the electronic pen **10A** from the posture detection circuit **24** of the digitizer **20A** and supplies the selected information to the associating circuit **415** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies within the spatial region of the position detection region DT and in the spatial region of the central region inside of the peripheral region PE, and selects the information regarding the relative pen posture from the posture conversion circuit **417** resulting from converting the pen posture of the electronic pen **10A** detected in the pen posture detection circuit **4102** and supplies the selected information to the associating circuit **415** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies in a spatial region outside thereof.

(95) Meanwhile, through the selection control signal SEA, the selection circuit **419** selects the information regarding the absolute pen posture to be detected in the spatial position detector **30**, which is supplied from the posture conversion circuit **417R** and results from converting the pen posture of the electronic pen **10A** detected in the posture detection circuit **24** of the digitizer **20A**, and supplies the selected information to the associating circuit **416** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies within the spatial region of the position detection region DT and in the spatial region of the central region inside of the peripheral region PE, and selects the information regarding the pen posture of the electronic pen **10A** from the pen posture detection circuit **4102** and supplies the selected information to the associating circuit **416** when the position (i.e., the position of the pen point) indicated by the electronic pen **10A** lies in a spatial region outside thereof.

(96) An example of a flow of a processing operation performed in the case where the input information processing circuit **41A** of the spatial position indication system according to this second embodiment is formed by a computer will now be described below with reference to FIG. 7 and FIG. 8, which follows FIG. 7. Note that, in this case, the various blocks of the input information processing circuit **41A** illustrated in FIG. 4 are functional software circuits implemented by a program for the computer.

(97) That is, the input information processing circuit **41A** determines from a signal from the digitizer **20A** whether or not a state in which the position indicated by the electronic pen **10A** is to be detected by the digitizer **20A** has been brought about (S1). When it is determined at S1 that the state in which the position indicated by the electronic pen **10A** is to be detected by the digitizer **20A** has not been brought about, the input information processing circuit **41A** determines whether or not the electronic pen **10A** can be detected by the spatial position detector **30** (S2). When it is determined at S2 that the electronic pen **10A** cannot be detected by the spatial position detector **30**, the input information processing circuit **41A** returns control to S1.

(98) When it is determined at S1 that the state in which the position indicated by the electronic pen **10A** is to be detected by the digitizer **20A** has been brought about, the input information processing circuit **41A** acquires the coordinates of the position indicated by the electronic pen **10A** and detected by the digitizer **20A** (S3), and determines whether or not the acquired coordinates of the indicated position are within the peripheral region PE (S4).

(99) When it is determined at S4 that the acquired coordinates of the indicated position are not within the peripheral region PE but within the region inside thereof, the input information processing circuit **41A** associates the information regarding the pen posture and the information regarding the position indicated by the electronic pen **10A** from the digitizer **20A** with each other, and outputs the resultant to the display image generation circuit **42** for drawing image generation (S5).

(100) Next, the input information processing circuit **41A** subjects the information regarding the position indicated by the electronic pen **10A** and detected in the spatial position detection circuit **4101** of the spatial information processing circuit **410** to coordinate conversion to be converted to information regarding the coordinate system shared with the digitizer **20A** (S6). Then, the input information processing circuit **41A** associates the information regarding the position indicated by the electronic pen **10A** subjected to the coordinate conversion, and the information regarding the pen posture detected in the pen posture detection circuit **4102** of the spatial information processing circuit **410**, with each other, and outputs the resultant to the display image generation circuit **42** for VR image generation (S7). After S7, the input information processing circuit **41A** returns control to S1, and the acts subsequent to S1 are repeated.

(101) Next, when it is determined at S4 that the acquired coordinates of the indicated position are within the peripheral region PE, or when it is determined at S2 that the electronic pen **10A** can be detected by the spatial position detector **30**, the input information processing circuit **41A** subjects the information regarding the position indicated by the electronic pen **10A** and detected in the spatial position detection circuit **4101** of the spatial information processing circuit **410** to coordinate conversion to be converted to information regarding the coordinate system shared with the digitizer **20A** (S11 in FIG. 8). In addition, the input information processing circuit **41A** acquires the information regarding the posture of the digitizer **20A** detected in the digitizer posture detection circuit **4103** of the spatial information processing circuit **410** (S12).

(102) Next, the input information processing circuit **41A** converts the information regarding the pen posture from the posture detection circuit **24** of the digitizer **20A** to information regarding an absolute posture in a spatial region, instead of a posture relative to the digitizer **20A**, using the information regarding the posture of the digitizer **20A** acquired at S12 (S13). Then, the input information processing circuit **41A** associates the information regarding the position indicated by the electronic pen **10A** resulting from the coordinate conversion at S11 and the information

regarding the absolute posture obtained at S13 with each other, and outputs the resultant to the display image generation circuit 42 for drawing image generation (S14).

(103) Next, the input information processing circuit 41A converts the information regarding the pen posture from the pen posture detection circuit 4102 of the spatial information processing circuit 410 to information regarding a posture relative to the digitizer 20A, using the information regarding the posture of the digitizer 20A acquired at S12 (S15). Then, the input information processing circuit 41A associates the information regarding the position indicated by the electronic pen 10A resulting from the coordinate conversion at S11 and the information regarding the absolute posture obtained at S13 with each other, and outputs the resultant to the display image generation circuit 42 for VR image generation (S16). After S16, the input information processing circuit 41A returns control to S1, and the acts subsequent to S1 are repeated.

(104) The input information processing circuit 41A of the spatial position indication system according to this second embodiment has the above-described configuration, and thus, as the information regarding the position indicated by the electronic pen 10A in each of the 3D drawing-related process and the VR image-related process, the information regarding the position indicated by the electronic pen 10A and detected in the position detection circuit 23 of the digitizer 20A with higher precision than when detected in the spatial position detector 30 is used for the central region inside of the peripheral region PE of the input surface 21S within the spatial region of the position detection region DT of the digitizer 20A, and the information regarding the position indicated by the electronic pen 10A and detected in the spatial position detection circuit 4101 of the spatial position detector 30 with higher precision than when detected in the position detection circuit 23 of the digitizer 20A is used for the peripheral region PE (including a corresponding space over the input surface 21S) of the input surface 21S within the spatial region of the position detection region DT of the digitizer 20A and a spatial region outside of the spatial region of the position detection region DT.

(105) In addition, as the information regarding the pen posture of the electronic pen 10A in the 3D drawing-related process, the information regarding the pen posture of the electronic pen 10A detected in the posture detection circuit 24 of the digitizer 20A with higher precision than when detected in the spatial position detector 30 is used for the central region inside of the peripheral region PE of the input surface 21S within the spatial region of the position detection region DT of the digitizer 20A, and the information regarding the pen posture of the electronic pen 10A detected in the pen posture detection circuit 4102 of the spatial position detector 30 with higher precision than when detected in the position detection circuit 23 of the digitizer 20A is used for the peripheral region PE (including the corresponding space over the input surface 21S) of the input surface 21S within the spatial region of the position detection region DT of the digitizer 20A and the spatial region outside of the spatial region of the position detection region DT.

(106) Accordingly, both the 3D drawing-related process and the VR image-related process can be performed always using information with high precision in this second embodiment as in the first embodiment. In this case, as in the first embodiment, advantageous convenience is provided in that a complementary use of the information detected in the digitizer 20A and the information detected in the spatial position detector 30 is possible in each of the 3D drawing-related process and the VR image-related process.

Third Embodiment

(107) While, in each of the first embodiment and the second embodiment described above, the input device according to an embodiment of the disclosure is applied to the spatial position indication system, an input device according to an embodiment of the disclosure is applicable also in the case where a position indication by an electronic pen is detected within a position detection region DT, including a hover region, in a tablet device provided with a digitizer.

(108) FIG. 9 is a diagram illustrating an outline of a system including an input device according to this third embodiment. In FIG. 9, the input device according to the third embodiment is made up of

an electronic pen **10B** and a tablet device **60** provided with a digitizer. As illustrated in FIG. 9, the tablet device **60** of this input device is, for example, placed on a desk **90**, and is connected to a computer **80**. In addition, in this example, the tablet device **60** is mounted on a mounting surface of the desk **90**, for example, at an inclination angle θ with respect thereto.

(109) In addition, in this third embodiment, a gyro sensor device **70** for detecting the pen posture of the electronic pen **10B** in a three-dimensional space over an input surface **61S** of the digitizer contained in the tablet device **60** is removably attached to the electronic pen **10B** on a rear-end side opposite to a pen-point side thereof. Note that the gyro sensor device **70** may be contained in the electronic pen **10B** on the rear-end side opposite to the pen-point side thereof.

(110) FIG. **10** is a block diagram for explaining an example functional configuration of the electronic pen **10B**, the tablet device **60**, and the gyro sensor device **70** of the input device according to this third embodiment. In this case, the gyro sensor device **70** is configured to include, for example, a triaxial gyro sensor **71** and a wireless communication circuit **72**. The wireless communication circuit **72** is formed by, for example, short-range wireless communication device that complies with the Bluetooth (registered trademark) standard. The wireless communication circuit **72** is not limited to the above, but may alternatively be configured to perform optical communication, such as infrared communication.

(111) In this example, the gyro sensor device **70** is attached to a housing of the electronic pen **10B**, and thus, the triaxial gyro sensor **71** outputs an output signal according to an inclination angle, an inclination direction, and a rotation angle of the electronic pen **10B** in the three-dimensional space. Here, the output signal of the electronic pen **10B** according to the inclination angle, the inclination direction, and the rotation angle of the electronic pen **10B** detected in the triaxial gyro sensor **71** does not correspond to the posture of the electronic pen **10B** relative to the input surface **61S** of the tablet device **60**, but corresponds to the absolute posture thereof in the three-dimensional space based on the axis of the earth. The wireless communication circuit **72** transmits the output signal from the triaxial gyro sensor **71** to the tablet device **60** in a wireless manner.

(112) The electronic pen **10B** includes an interaction circuit **100** that interacts with a sensor **61** of the tablet device **60**. The interaction circuit **100** is coupled to the sensor **61** of the tablet device **60** by an electromagnetic induction method or a capacitive coupling method to exchange signals therewith. The interaction circuit **100** of the electronic pen **10B** in this example exchanges signals for position detection of the position indicated by the electronic pen **10B** with the sensor **61**, and does not exchange therewith signals for detecting the pen posture, such as the inclination angle, inclination direction, rotation angle

(113) In the tablet device **60**, from signals obtained as a result of an interaction between the sensor **61** and the electronic pen **10B**, the position indicated by the electronic pen **10B** is detected in an indicated position detection circuit **62**. Information about the position indicated by the electronic pen **10B** detected in the indicated position detection circuit **62** is supplied to an associating circuit **63**.

(114) The tablet device **60** according to this third embodiment includes a wireless communication circuit **64** that performs wireless communication with the wireless communication circuit **72** of the gyro sensor device **70**. A detection output of the triaxial gyro sensor **71** of the gyro sensor device **70** received by this wireless communication circuit **64** is supplied to an electronic pen posture detection circuit **65**, and the pen posture of the electronic pen **10B** is detected. Information about the pen posture of the electronic pen **10B** detected in this electronic pen posture detection circuit **65** is supplied to a relative posture calculation circuit **66**.

(115) The tablet device **60** according to this embodiment includes a gyro sensor **67** for detecting the inclination angle and inclination direction of the tablet device **60**. A sensor output of this gyro sensor **67** is supplied to a tablet posture detection circuit **68**. The tablet posture detection circuit **68** detects the inclination angle and inclination direction of the tablet device **60** on the basis of the sensor output of the gyro sensor **67**, and supplies information (i.e., information regarding a tablet

posture) about the detected inclination angle and inclination direction of the tablet device **60** to the relative posture calculation circuit **66**.

(116) The relative posture calculation circuit **66** corrects the information regarding the absolute pen posture of the electronic pen **10B** in the three-dimensional space from the electronic pen posture detection circuit **65** using the information regarding the tablet posture from the tablet posture detection circuit **68**, thereby calculating the pen posture of the electronic pen **10B** relative to the input surface **61S** of the tablet device **60**. Then, the relative posture calculation circuit **66** supplies information regarding the calculated relative pen posture of the electronic pen **10B** to the associating circuit **63**.

(117) As is similarly the case with the associating circuits described above, the associating circuit **63** associates the information regarding the position indicated by the electronic pen **10B** and the information regarding the relative pen posture of the electronic pen **10B** with each other, and outputs the resultant to the computer **80**.

(118) As described above, in the input device according to this third embodiment, when a position indicated by a pen point of the electronic pen **10B** can be obtained, information regarding a pen posture of the electronic pen **10B** which is correspondingly detected in the three-dimensional space through the gyro sensor device **70** provided on the rear-end side of the electronic pen **10B** can be outputted as information regarding a pen posture corresponding to the position indicated by the electronic pen **10B**.

Other Embodiments or Modifications

(119) While, in each of the first embodiment and the second embodiment described above, the spatial position detector **30** is configured to include the light emitting/tracking devices that emit infrared laser beams and the trackers, it is needless to say that this configuration is not essential to the disclosure. For example, the spatial position detector may be configured to use other invisible light sensors, visible light sensors, any combination thereof, or the like.

(120) In addition, in each of the first embodiment and the second embodiment described above, the operator of the electronic pen **10** wears the HMD **50**, and is therefore not able to directly view the digitizer **20**. Accordingly, a virtual image of the digitizer **20** may be drawn in an image of a virtual space displayed on the HMD **50** to enable the operator to recognize the position of the digitizer **20** through the display screen of the HMD **50**.

(121) Further, while, in each of the first embodiment and the second embodiment described above, the coordinate values of the spatial position detector **30** are converted to coordinate values in the spatial coordinate system of the digitizer **20** to correct an error in the two spatial coordinate systems, the coordinate values of the position detection region DT of the digitizer **20** may conversely be converted to coordinate values of the spatial position detector **30**. Note that the conversion between the two spatial coordinate systems is not essential but optional.

(122) While, in each of the first embodiment and the second embodiment described above, the electronic pen and the digitizer used employ the electromagnetic induction method, this is not essential to the disclosure, and an electronic pen and a digitizer that employ a capacitive method (e.g., an active capacitive coupling method, a passive capacitive coupling method, etc.) may naturally be used instead.

(123) Also note that each of the digitizers according to the first embodiment and the second embodiment and the tablet device according to the third embodiment may be a portable cellular phone terminal often called a smart phone.

(124) While it has been assumed that 3D images are drawn by the spatial position indication system according to each of the first embodiment and the second embodiment described above, the images to be drawn may alternatively be 2D (Two-Dimensional) drawing images or 2.5D drawing images.

DESCRIPTION OF REFERENCE SYMBOLS

(125) **10**, **10A**, **10B** . . . Electronic pen, **20**, **20A** . . . Digitizer, **21S** . . . Input surface, **22** . . . Sensor, **23** . . . Position detection circuit, **24** . . . Posture detection circuit, **30** . . . Spatial position detector,

40 . . . Spatial drawing information generation device, **41** . . . Input information processing circuit, **4101** . . . Spatial position detection circuit, **4102** . . . Pen posture detection circuit, **50** . . . HMD

Claims

1. A Virtual Reality (VR) image generation system comprising: an electronic device including a digitizer which, in operation, detects a position indicated by an electronic pen; and one or more processors; and one or more memory devices storing instructions that, when executed by the one or more processors, cause the one or more processors to: obtain an absolute posture of an electronic pen, obtain a posture of the digitizer of the electronic device, calculate a relative posture of the electronic pen with respect to a physical input surface of the electronic device based on the absolute posture of the electronic pen and the posture of the digitizer, generate a VR image of the electronic pen based on the relative posture of the electronic pen with respect to the physical input surface of the electronic device, and control a display to display the VR image of the electronic pen in a space of the VR.
2. The VR image generation system according to claim 1, wherein: the relative posture of the electronic pen with respect to the physical input surface of the electronic device is calculated based on a posture of the electronic pen detected by the electronic device.
3. The VR image generation system according to claim 1, wherein: the relative posture of the electronic pen with respect to the physical input surface of the electronic device is calculated based on a posture of the electronic pen detected by a gyro sensor of the electronic pen and a posture of the electronic device detected by a gyro sensor of the electronic device.
4. The VR image generation system according to claim 1, wherein: the posture of the digitizer includes an inclination angle of the digitizer.
5. The VR image generation system according to claim 4, wherein: the posture of the digitizer includes an inclination direction of the digitizer.
6. The VR image generation system according to claim 5, wherein: the absolute posture of the electronic pen includes an inclination angle of the electronic pen.
7. The VR image generation system according to claim 6, wherein: the absolute posture of the electronic pen includes an inclination direction of the electronic pen.
8. The VR image generation system according to claim 7, wherein: the absolute posture of the electronic pen includes a rotation angle of the electronic pen.
9. A Virtual Reality (VR) image generation method comprising: obtaining an absolute posture of an electronic pen, obtaining a posture of a digitizer of an electronic device that detects a position indicated by the electronic pen, calculating a relative posture of the electronic pen with respect to a physical input surface of the electronic device including the digitizer that detects the position indicated by the electronic pen, and generating a VR image of the electronic pen based on the relative posture of the electronic pen with respect to the physical input surface of the electronic device, and controlling a display to display the VR image of the electronic pen in a space of the VR.
10. The VR image generation method according to claim 9, wherein: the calculating includes calculating the relative posture of the electronic pen with respect to the physical input surface of the electronic device based on a posture of the electronic pen detected by the electronic device.
11. The VR image generation method according to claim 9, wherein: the calculating includes calculating the relative posture of the electronic pen with respect to the physical input surface of the electronic device based on a posture of the electronic pen detected by a first gyro sensor of the electronic pen and a posture of the electronic device detected a second gyro sensor of the electronic device.
12. The VR image generation method according to claim 9, wherein: the posture of the digitizer includes an inclination angle of the digitizer.

13. The VR image generation method according to claim 12, wherein: the posture of the digitizer includes an inclination direction of the digitizer.
 14. The VR image generation method according to claim 13, wherein: the absolute posture of the electronic pen includes an inclination angle of the electronic pen.
 15. The VR image generation method according to claim 14, wherein: the absolute posture of the electronic pen includes an inclination direction of the electronic pen.
 16. The VR image generation method according to claim 15, wherein: the absolute posture of the electronic pen includes a rotation angle of the electronic pen.
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